

COGASet: A Persona-Driven Dataset and Evaluation Framework for Cognitively Accessible LLM-Generated Content

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Abstract

While the Web Content Accessibility Guidelines (WCAG) benefit from mature automated testing tools, cognitive accessibility evaluation under the World Wide Web Consortium (W3C) Cognitive and Learning Disabilities Accessibility Task Force (COGA) guidelines lacks standardized assessment instruments. This paper presents COGASet, a dataset of 60 Large Language Model (LLM) generated product information passages spanning 20 cognitive accessibility personas (including ADHD, dyslexia, aphasia, autism spectrum, anxiety, intellectual disability, and age-related cognitive changes). Passages were generated by four major LLM families (ChatGPT, Claude, DeepSeek, and Gemini). Each persona received three tailored prompting strategies generated by ChatGPT: marketing-style Baseline, COGA Zero-shot (rules only), and COGA Few-shot (rules with exemplars). We operationalized six COGA design patterns through a human evaluation framework: purpose-first structure, plain language, calm tone, descriptive action cues, help clarity, and overall ease. Twenty trained evaluators independently assessed all the passages. COGA-aware prompting showed substantial improvements over baseline across multiple dimensions: Zero-shot achieved 92.5% “easy to understand” ratings versus 55% for Baseline, with gains in plain language (60%→97.5%), calm tone (77.5%→100%), and help clarity (60%→95%). Unexpectedly, Zero-shot outperformed Few-shot on calm tone (100% vs. 87.5%, $p=.022$), suggesting that explicit rules may be more effective than examples for certain accessibility requirements. Inter-rater reliability was adequate (average agreement 83.92%, dimension-specific 72.55-92.16%). We release COGASet and our evaluation framework to support future research in cognitive accessibility.

CCS Concepts

• **Human-centered computing**; • **Accessibility**; **Empirical studies in Human-computer interaction (HCI)**; • **Computing methodologies**; • **Natural language generation**;

Keywords

Cognitive accessibility, COGA guidelines, Accessibility dataset, Large Language Models, Prompt engineering, Personas, Accessible content generation



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1 Introduction

Large language models (LLMs) are increasingly integrated into web content creation workflows [1–3], with direct implications for digital accessibility. While the Web Content Accessibility Guidelines (WCAG) [4] benefit from mature automated testing tools such as AXE and WAVE [5, 6], cognitive accessibility under the W3C Cognitive and Learning Disabilities Accessibility Task Force (COGA) guidelines [7] presents fundamentally different challenges. Unlike WCAG’s testable criteria (e.g., “contrast ratio of at least 4.5:1”), COGA recommendations require human judgment about vocabulary, sentence structure, tone, and cognitive load. No standardized evaluation instruments currently exist [7, 8], leaving content creators unable to systematically verify COGA compliance and researchers without benchmarks to assess model performance across diverse cognitive profiles.

This paper makes two contributions. First, COGASet: a dataset of 60 LLM-generated product information passages spanning 20 cognitive accessibility personas (ADHD, dyslexia, aphasia, autism spectrum, anxiety, intellectual disability, and others), generated by four LLM families using three prompting strategies. Second, a Human Evaluation Framework operationalizing six core COGA design patterns with structured rating scales. We investigate: (RQ1) Does COGA-aware prompting improve text generation across cognitive accessibility dimensions? (RQ2) Does Few-shot prompting outperform Zero-shot? (RQ3) Can the evaluation framework reliably distinguish accessibility quality across LLM outputs?

2 Related Work

Recent work has established various benchmarks for evaluating LLMs in web development contexts. WebGen-Bench introduced a comprehensive framework for assessing LLMs’ ability to generate interactive and functional websites from scratch, focusing on technical correctness and functionality [9]. Similarly, Web2code presented a large-scale dataset pairing webpages with their source code, enabling evaluation of multimodal LLMs on code generation tasks [10]. These efforts demonstrate the growing need for systematic evaluation of LLM outputs in web contexts.

In the domain of automated testing, researchers have explored LLMs’ capability to generate web-form tests, examining their effectiveness in creating functional test cases [11]. Comparative

studies have investigated how different LLMs perform on web development code generation tasks, revealing significant variation in output quality across models and prompting strategies [12]. While these benchmarks assess technical and functional dimensions of LLM-generated web content, they do not address accessibility concerns.

The intersection of LLMs and accessibility has received limited attention. Recent work investigated whether ChatGPT generates accessible code, examining WCAG compliance in LLM-generated source code and identifying systematic accessibility violations [13]. The W3C COGA guidelines address content structure, language clarity, and cognitive load, but lack standardized evaluation instruments. Our work addresses this gap by introducing COGASet, the first dataset for assessing cognitive accessibility in LLM-generated content across diverse cognitive profiles, alongside a human evaluation framework operationalizing COGA design patterns.

3 COGASet Dataset and Evaluation Framework

3.1 Personas and Content Domain

We selected e-commerce product information (online laptop store) as our test domain: shopping is a COGA priority task [14] and the structured format (page title, introduction, product descriptions, help section) enables direct comparisons. We developed 20 cognitive accessibility personas from COGA user stories and personas [7], spanning attention and executive function (ADHD, easily distracted), reading and language (dyslexia, aphasia, low reading level), processing and memory (dyscalculia, working memory difficulty), anxiety (commitment anxiety, credential worry, decision paralysis, time-pressure sensitivity), neurodivergence (autism spectrum), age and injury (older adults, post-TBI), and low digital literacy (see Appendix A1 for the complete set). These profiles are structured accessibility-need mappings derived from COGA objectives, prioritizing cognitive diversity over biographical depth.

3.2 Prompting Strategies and Generation

Each persona received three prompting strategies generated by ChatGPT (GPT-4): (1) Baseline: marketing-style prompts requesting “energetic, persuasive, tech-savvy” copy with persona description but no COGA guidance; (2) COGA Zero-shot: explicit COGA-aligned rules tailored per persona, including core rules (clear purpose statement, short sentences with common words, no urgency language, simple help path) and persona-specific guidance (e.g., for aphasia: “repeat the noun instead of pronouns”; for dyscalculia: “use words for comparisons instead of dense numbers”); (3) COGA Few-shot: same rules as Zero-shot plus 2-3 BAD/GOOD exemplar pairs demonstrating violations versus COGA compliance. Passages were generated by four LLM families assigned to persona groups of five (GPT-4, Claude Sonnet 3.5, DeepSeek, Gemini 1.5 Pro), yielding 60 unique passages. This design confounds persona and model effects; model comparisons should therefore be treated as descriptive.

3.3 Evaluation Framework and Procedure

We operationalized six COGA design patterns as assessable dimensions, each with a structured rating scale and evidence requirements (Table 1). Twenty senior IT students, trained in W3C COGA

guidance and blinded to prompting condition and LLM source, independently rated all 60 passages in randomized order, with two evaluators per persona. Inter-rater reliability was computed as percentage agreement per dimension. Using IT students rather than accessibility specialists reduces the personal bias domain experts may introduce from their own lived experiences with cognitive disabilities.

4 Results

4.1 RQ1: Effect of COGA-Aware Prompting

Table 2 reports the percentage rated CLEAR (or EASY for overall ease) by dimension and condition ($n=40$ evaluations per cell). COGA-aware prompting showed substantial improvements over Baseline across all dimensions. Overall ease increased from 55% (Baseline) to 92.5% (Zero-shot) and 90% (Few-shot), a gain of 37.5 percentage points. The largest improvements were in plain language (60%→97.5%), help clarity (60%→95%), and calm tone (77.5%→100%). Purpose-first structure (82.5% Baseline) and descriptive actions (80% Baseline) were already relatively high, suggesting marketing prompts naturally produce some structural clarity. Kruskal-Wallis tests confirmed significant differences for plain language ($H=29.17$, $p<.001$), overall ease ($H=21.14$, $p<.001$), help clarity ($H=15.13$, $p<.001$), and calm tone ($H=9.78$, $p=.008$); purpose-first structure and descriptive actions were not significant.

4.2 RQ2: Zero-shot vs. Few-shot

Contrary to typical prompt engineering findings [15], Zero-shot outperformed Few-shot on calm tone (100% vs. 87.5%, $p=.022$) and showed an advantage on help clarity (95% vs. 82.5%, $p=.080$); both conditions were equal on plain language (97.5%) as shown in Table 3. We attribute this to two mechanisms: first, COGA rules are sufficiently precise that exemplars add noise rather than clarity (e.g., “avoid urgency terms: hurry, last chance, limited time” is unambiguous). Second, the BAD/GOOD exemplar structure may prime a two-stage generation process where models produce marketing-style content (primed by BAD examples) then attempt COGA transformation, leaving residual pressure cues. For accessibility requirements demanding consistency, concise rule-based Zero-shot prompting may be more effective than exemplar augmentation.

4.3 RQ3: Framework Reliability

Overall inter-rater agreement (Table 4) was 83.92%. Dimension-specific agreement ranged from 72.55% (overall ease) to 92.16% (help clarity): descriptive actions 88.24%, calm tone 86.27%, plain language 84.31%, purpose-first 78.43%. Dimensions tied to directly observable textual features showed higher agreement than those requiring holistic judgment. These results confirm that systematic human assessment of COGA cognitive accessibility is feasible with adequate reliability.

5 Discussion and Conclusion

This paper presents COGASet, addressing a critical gap in cognitive accessibility evaluation. Our results indicate that LLMs can generate cognitively accessible content through prompt engineering alone, without model fine-tuning. The 55% to 92.5% improvement in

Table 1: Evaluation framework dimensions

Dimension	COGA Rationale	Operational Definition	Rating Scale	Evidence Requirement
1. Purpose-First Structure	Users with cognitive disabilities benefit from immediately understanding what a page is for [1, Obj. 1]	Whether opening sentences clearly state the page’s purpose	CLEAR: Purpose stated in first sentence PARTIAL: Purpose stated by second sentence NOT CLEAR: Purpose absent or buried deeper	Quote first 1-2 sentences
2. Plain Language	Everyday vocabulary and short sentences reduce cognitive load [1, Obj. 3]	Use of familiar words and sentences ≤ 20 words with one idea each	CLEAR: Consistently plain language throughout MIXED: Some plain language but also complex sentences or jargon NOT CLEAR: Predominantly complex or technical language	Provide example sentence
3. Calm Tone	Pressure language increases anxiety and reduces task completion [1, Obj. 7]	Absence of urgency/pressure terms (hurry, last chance, limited time, act now, flash sale, expires/expiring, don’t miss out, now or never)	CLEAR: No pressure language MINOR: 1-2 instances of pressure terms NOT CLEAR: 3+ instances or pervasive urgency	List matched terms with brief context
4. Descriptive Action Cues	Clear labels specifying both action and target help users predict outcomes [WCAG 2.4.6, 3.3.2; 1, Obj. 6]	Button/link labels that state the action and target (e.g., “Add this laptop to cart” vs. “Click here”)	CLEAR: 3+ descriptive action labels PARTIAL: 1-2 descriptive labels NOT CLEAR: Mostly vague or ambiguous labels	List action phrases found
5. Help Clarity	Simple help paths with clear next-step explanations support users with memory, executive function, and anxiety challenges [1, Obj. 5]	Help section provides (a) simple steps to get help, and (b) explanation of what happens next	CLEAR: Both steps and outcomes explained PARTIAL: Steps present but outcomes vague NOT CLEAR: Help absent or unclear	Paste help snippet and enumerate steps
6. Overall Ease	Holistic assessment of cognitive accessibility	Evaluator’s judgment of whether the passage would be easy for the target persona to understand and use	EASY: Clear and accessible throughout OKAY: Acceptable with some issues HARD: Difficult to understand or use	Brief justification (1-2 sentences)

Table 2: Percentage of passages rated CLEAR/EASY by dimension and prompting condition (n=40 evaluations per cell)

Dimension	Baseline	Zero-shot	Few-shot
Purpose-first	82.5%	92.5%	85.0%
Plain language	60.0%	97.5%	97.5%
Calm tone	77.5%	100.0%	87.5%
Descriptive actions	80.0%	90.0%	92.5%
Help clarity	60.0%	95.0%	82.5%
Overall ease	55.0%	92.5%	90.0%

overall ease (Zero-shot) demonstrates that COGA-aware prompting achieves substantial gains in existing workflows. Persona-specific rules proved valuable: COGA design patterns translate effectively to prompt instructions because they specify observable qualities (short sentences, calm tone, explicit help paths) rather than abstract

goals. The unexpected Zero-shot advantage over Few-shot on calm tone challenges conventional prompt engineering wisdom and suggests that highly specific accessibility requirements may not benefit from exemplar augmentation.

Table 3: Statistical comparisons across conditions

Dimension	Zero-shot	Few-shot	Difference	p-value
Purpose-first	92.5%	85.0%	+7.5 points	n.s.
Plain language	97.5%	97.5%	0	n.s.
Calm tone	100.0%	87.5%	+12.5 points	p=.022*
Descriptive actions	90.0%	92.5%	-2.5 points	n.s.
Help clarity	95.0%	82.5%	+12.5 points	p=.080
Overall ease	92.5%	90.0%	+2.5 points	n.s.

* p<.05, n.s. = not significant

Table 4: Average agreement across the six rating dimensions

Dimension	Agreement
Purpose-first	78.43%
Plain language	84.31%
Calm tone	86.27%
Descriptive actions	88.24%
Help clarity	92.16%
Overall ease	72.55%

Several limitations constrain generalizability. Evaluation was conducted by IT students, not end users with cognitive disabilities; end-user validation is essential. The study focused on e-commerce content and requires extension to healthcare, government, and educational domains. The 20 personas, while diverse, omit severe intellectual disabilities, co-occurring conditions, and combinations of cognitive and sensory disabilities. The confounded persona-model design limits definitive model comparisons. Future work should include factorial designs separating persona and model effects, validation with target users, semi-automated evaluation tools, and multilingual adaptation. COGASet and our evaluation framework are released to support this research agenda.

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A APPENDICES

A.1 Complete Personas Set

Table 5: COGASet personas and their cognitive accessibility needs

ID	Persona	Primary Need	COGA Objectives
1	Overwhelmed by Urgency	Calm tone, no pressure language	Obj 7
2	Working Memory Difficulty	Explicit steps, undo options	Obj 6
3	Low Digital Literacy	Everyday words, concrete examples	Obj 1
4	Dyslexia	Short sentences, no ALL CAPS	Obj 3
5	ADHD / Easily Distracted	Scannable chunks, one action per section	Obj 7
6	Autism / Literal Language	No idioms/metaphors, consistent structure	Obj 4
7	Aphasia	Very short sentences, repeated nouns	Obj 2
8	Dyscalculia	Word comparisons, minimal numbers	Obj 1
9	Executive Function	Explicit goals, checklists	Obj 8
10	Anxiety (Commitments)	Return reassurance, no pressure	Obj 7
11	Older Adult (Mild Cognitive)	Plain language, repeated actions	Obj 1
12	Post-TBI (Fatigue)	Short blocks, predictable order	Obj 6
13	Intellectual Disability (Mild)	Very plain language, repetition	Obj 1
14	ESL / Newcomer to English	No idioms, everyday vocabulary	Obj 1
15	Low Reading Level Teen	Grade 6-7 level, direct voice	Obj 1
16	Caregiver (Split Attention)	Quick summary, save/return option	Obj 8
17	Time-Pressure Sensitivity	No timeouts, explicit "no rush"	Obj 7
18	Account/Credential Anxiety	Privacy reassurance, simple recovery	Obj 7
19	Decision Paralysis	Few options, safe default marked	Obj 8
20	Sensory/Word Overload	Minimal adjectives, plain verbs	Obj 1